Signal, Image and Video Processing (2024) 18:8159–8167
https://doi.org/10.1007/s11760-024-03458-w

ORIGINAL PAPER

YOLO-GTWDNet: a lightweight YOLOv8 network with GhostNet backbone and transformer neck to detect handheld weapons for smart city applications

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Received: 12 April 2024 / Revised: 9 July 2024 / Accepted: 22 July 2024 / Published online: 31 July 2024
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Abstract
Weapon detection is the process of identifying handheld weapons such as guns, knives, etc., and creating a bounding box around them to highlight the spatial locations. Weapon detection is one of the key building blocks of the intelligent video surveillance system for security applications in smart cities. However, detecting handheld weapons from surveillance videos is quite challenging due to small object size, occlusion, illumination variation, model complexity, and latency. Hence, an efficient, novel, robust, and lightweight YOLOv8-based weapon detector with GhostNet backbone and C3 module with transformer block (C3TR) neck (YOLO-GTWDNet model) is proposed for detecting the weapons either from stored images or from the live video streams. The proposed model is trained using a weapon dataset named "Weapon7," which is developed by collecting various weapon classes, such as Axe, Bow and arrow, Gun, Knife, Lathi, Pistol, and Sword, from various publicly available datasets, Internet, and own camera capture. Extensive experimental analysis is carried out to demonstrate the effectiveness of the proposed YOLO-GTWDNet model. The proposed model outperforms the state-of-the-art models when compared using both quantitative and qualitative performance metrics. The deployment of the proposed model is expected to bolster public safety significantly, providing city authorities with a powerful tool to mitigate risks and swiftly address potential threats.

Keywords C3TR · Deep learning · GhostNet · Weapon detection · Weapon7 dataset · YOLOv8 · YOLO-GTWDNet model

1 Introduction

The global crime rate is increasing day by day, mostly due to the widespread use of portable weapons in violent incidents. Detection of handheld weapons by observing the CCTV monitors by the security personnel manually is one of the tedious, time-consuming, and erroneous tasks [20]. Hence, these systems can be enhanced to automatically detect handheld weapons during violent activity or before the actual happening of the violent activity by deploying a deep learning (DL)-based weapon detector inside the surveillance system. Automatic weapon detection can be treated as a subset of the object detection problem, where the targeted object to be detected is the weapon. Usually, weapon detectors achieve high detection accuracy at the cost of computational complexity (processing speed). But, weapon detectors should possess high inference speed (Frames Per Second, i.e., FPS) with competitive accuracy. Hence, there is always a trade-off between the desired processing speed and detection accuracy for the targeted application with available computational resources. Most of the previous works regarding weapon detection are targeted at concealed weapon detection using X-ray or millimeter wave images. There is not much work dedicated to weapon detection from surveillance videos. Few works based on traditional machine learning-based models [4, 23] related to weapon detection have been proposed. However, this article is focused on the development of a

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deep learning-based lightweight weapon detection model with competitive detection accuracy.

1.1 Contribution

The main contributions are as follows.

- A weapon dataset named "Weapon7" is developed by collecting various weapon classes, such as Axe, Bow and arrow, Gun, Kinfe, Lathi, Pistol, and Sword, from various publicly available datasets, Internet, and own camera capture.
- An efficient, novel, robust, and lightweight YOLOv8-based weapon detector with GhostNet backbone and C3TR neck (YOLO-GTWDNet model) is proposed for detecting weapons from stored images or live video streams.
- An end-to-end pipeline for detecting hand-held weapons from the live video stream is proposed.
- Extensive experimental analysis is carried out to demonstrate the effectiveness of the proposed YOLO-GTWDNet model.

1.2 Organization

The rest of the article is organized as follows. Related works have been briefly explained in Sect. 2. Problem statement is outlined in Sect. 3. The methodology used to develop the proposed YOLO-GTWDNet model is presented in Sect. 4. Subsequently, the experiments in terms of implementation details, results, and comparative analysis are presented in Sect. 5. Finally, the article is concluded in Sect. 6.

2 Related works

Recently, a few DL-based approaches have been used in the detection of handheld weapons such as Gun, Lathi, Swords, etc., from the real-time video stream. Broadly, these approaches can be divided into two categories such as single-stage weapon detectors and two-stage weapon detectors.

2.1 Single-stage weapon detectors

Single-stage weapon detectors perform weapon classification and bounding-box regression directly without using pre-generated region proposals. Single-stage weapon detectors are mostly based on the YOLO series and SSD series models. Detection of various handheld weapons such as Gun, Pistol, and screwdriver using YOLOv3 [16], CNN-based SSD [9], YOLOv4 [7], YOLOv5 [17], YOLOv7 [15], and YOLOv8 [12] have been proposed. However, all these models suffer from high model complexity which restricts the corresponding model deployment at the edge devices.

2.2 Two-stage weapon detectors

Two-stage weapon detectors perform weapon detection in two sequential stages: region proposal generation and object classification for each region proposal. Detection of various handheld weapons such as Gun, Pistol, knife, and screwdriver using Faster R-CNN [9, 18], Faster R-CNN Inception-v2 model [3, 8], mask R-CNN [13], and Inception-RestNetv2 [1] have been proposed. These models require more inference time and hence are not suitable for real-time applications.

Further, the reported works are limited to only one to three classes of weapons. Most of them are only transfer-learning-based models. Hence, though multiple works have been proposed, there is still ample scope to develop an efficient and robust DL model to detect multiple classes of hand-held weapons for real-time applications for larger classes of weapons.

3 Problem statement

Usually, high detection accuracy is achieved at the cost of computational complexity, and hence, weapon detectors with high computational complexity are not suitable for real-time applications. Further, weapon detectors should possess high inference speed with competitive detection accuracy for real-time applications. Hence, the proposed YOLO-DTWDNet attempts to achieve high processing speed with competitive accuracy. Given video frames f_i or image I, the DL-based lightweight weapon detector "YOLO-GTWDNet" trained on "Weapon7" dataset can detect (both classification and localization) weapons with class labels from any one of the trained classes list "Axe (0), Bow and arrow (1), Gun (2), Kinfe (3), Lathi (4), Pistol (5), and Sword (6)," and generate bounding box BB around the detected weapon for providing localization information in spatial domain.

4 Proposed methodology

Ultralytics has YOLOv8 [10] using Pytorch to optimize the accuracy and efficiency of object detection. Similar to YOLOv5 [11], YOLOv8 is also available in five different model scales: YOLOv8n (nano), YOLOv8s (small), YOLOv8m (medium), YOLOv8l (large), and YOLOv8x (extra-large). The detection accuracy of YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x improves in sequential order at the expense of increased computing cost due to larger models. Here, YOLOv8x is considered as the base model due to its high detection accuracy. However, the computational complexity of the YOLOv8x is maximum, which

results in its slow processing speed. Hence, the complexity of the YOLOv8x model can be reduced by suitable architectural modifications without compromising on the detection accuracy. Hence, the proposed YOLO-GTWDNet model uses GhostNet [5] as its backbone and C3 module with transformer block (C3TR) in the neck to achieve lightweight of the model without compromising on the detection accuracy. Here, the C3 module means a cross-stage parallel network (CSP) bottleneck with three convolutions. The network architecture of the proposed YOLO-GTWDNet model is presented in Fig. 1.

4.1 Improvements introduced in backbone section of the proposed YOLO-GTWDNet model

Usually, deep neural networks (DDNs) utilize deeper architecture with more parameters to better fit large-scale datasets. Recently, GhostNet [5, 6, 22] which aimed at generating more number of features with cheaper operations by avoiding redundant parameters is proposed. In the proposed YOLO-GTWDNet model, the complex backbone is replaced with simple GhostNet to reduce the overall model complexity without affecting the detection accuracy. The GhostNet model [5] is constructed by employing the Ghost module, which creates a greater number of features while consuming fewer parameters through the use of a more cost-effective procedure known as Ghost Convolution. Ghost convolution is a technique that splits a regular convolutional layer in DDNs into two separate operations. The initial stage entails conventional convolution with precise regulation of the overall number of convolutions performed. The second phase involves doing a sequence of simple linear operations to produce additional feature maps from the inherent feature maps obtained in the first stage. Therefore, the GhostNet reduces the total number of parameters and computational difficulties without altering the size of the output feature map, in comparison to the general convolutional layer. This is achieved while maintaining equal recognition performances [5].

GhostNet attempts to reduce the widely existing redundancy in intermediate feature maps by reducing the number of convolution filters used to generate them. For the given input $X \in R^{c \times h \times w}$, Ghost features can be generated through a series of cheap linear operations on each intrinsic feature in Y' by using Eq. 1 [5].

$$y_{ij} = \Phi_{ij}\left(y'_i\right), \ \forall\, i = 1, 2, ..., m; \ \ j = 1, 2, ..., s \tag{1}$$

Here, y'_i is the ith intrinsic feature map in Y' and Φ_{ij} is the jth (except last one) linear operation for generating j^{th} ghost feature map y_{ij}. The last Φ_{ij}, i.e., for $j = s$, represents the identity mapping for preserving the intrinsic feature maps. The output of the Ghost module can be represented by $n = m.s$ feature maps as represented in Eq. 2 [5].

$$Y = [y_{11}, y_{12}, ..., y_{ms}] \tag{2}$$

Finally, Ghost convolution performed by the Ghost module achieved the model compression by using fewer convolution operations. The GhostNet backbone is realized with the help of cascaded connections GhostConv and C3Ghost blocks as shown in Fig. 1. GhostConv block performs Ghost convolution operation. C3Ghost acts as a C3 module with GhostBottleneck. The features resulting from the backbone are subsequently passed to the neck section for further fine-tuning.

4.2 Improvements introduced in neck section of the proposed YOLO-GTWDNet model

The neck section is an intermediate processing stage where the extracted features from the backbone are refined and given to the head section for final predictions. The spatial Pyramid Pooling Fast (SPPF) bottleneck is replaced by the Ghost Bottleneck in the proposed architecture to decrease computational complexity without significantly affecting detection accuracy. The Ghost Bottleneck refers to a skip connection block that is similar to the basic residual block found in ResNet. The residual block incorporates many convolutional layers and shortcuts. However, Ghost bottleneck stacks two ghost modules in a cascaded manner with a shortcut connection as shown in Fig. 2a [5]. The first and second Ghost modules act as expansion layers to increase the number of channels and decrease the number of channels for matching with the shortcut path, respectively. Expansion ratio refers to the ratio of the number of output channels to that of the input channels. Finally, a shortcut path is connected between the inputs and outputs of these two Ghost modules. A transformer layer extracts more efficient features by modeling the relationship among all pixels through an attention mechanism to capture global context information [21]. Further, this layer can perform complementary to the convolutional layer. Hence, the last C2f block is replaced by the C3TR due to its improved feature representation, reduced computational complexity, residual connections, flexibility in feature fusion, regularization, and feature reuse properties. The C3TR block incorporates transformer structure and ViT structure [25] into YOLOv8 architecture as shown in Fig. 2b. The C3TR module partitions the input image into image blocks of a specific size using the Patch Embedding operation. These separated image blocks are then merged into a sequence and passed on to the Transformer Encoder for extracting features.

4.3 Head section of the proposed YOLO-GTWDNet model

The head section of the proposed YOLO-GTWDNet model is realized by using Detect blocks of the YOLOv8 archi-

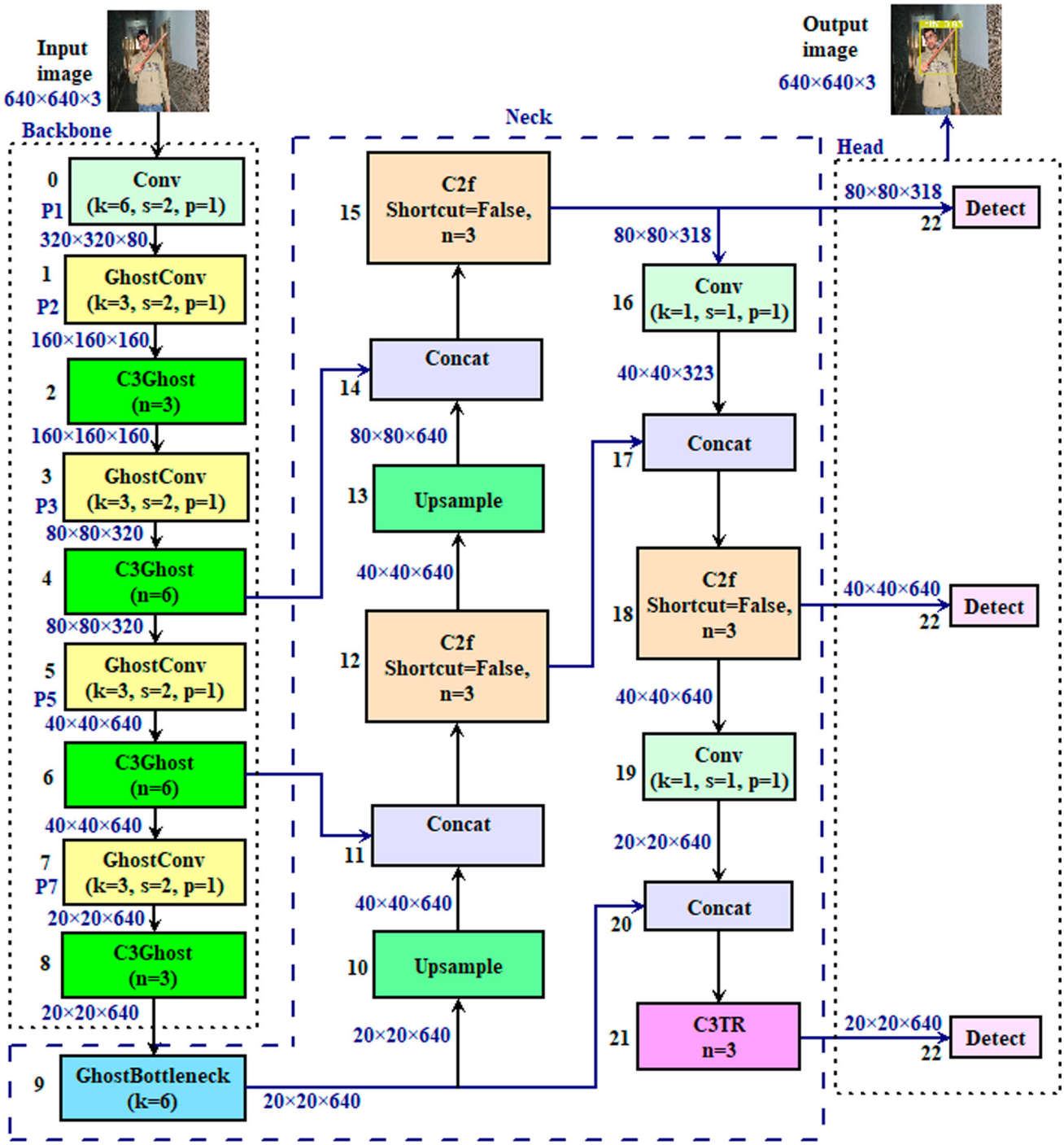

Fig. 1 Network architecture of the proposed YOLO-GTWDNet model

tecture. The detect block consists of two parallel streams of cascaded convolution layers corresponding to bounding box loss and classification loss. The upper stream corresponding to the bounding box losses is responsible for predicting the bounding box around the object. Similarly, the lower stream corresponding to the classification loss predicts classes with confidence level.

5 Experimental analysis

A comprehensive experimental analysis of the proposed framework is carried out as follows.

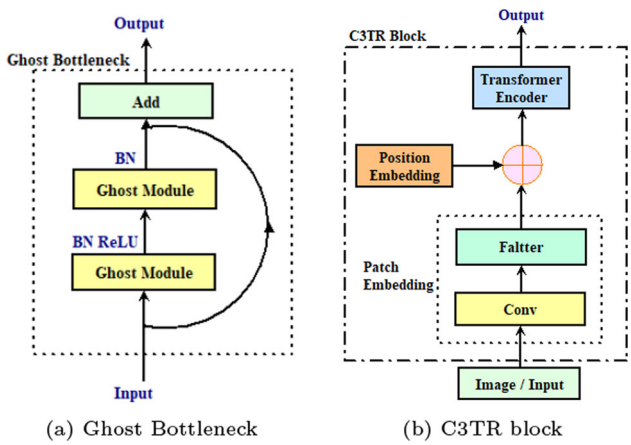

Fig. 2 Structures of the Ghost Bottleneck (with stride=1) and C3TR blocks

5.1 Experimental setup

The experimental setup comprises hardware and software details used to develop the proposed YOLO-GTWDNet model, is presented in Table 1.

5.2 Dataset

Images of the seven different types of most frequently used weapons such as Axe (0), Bow and arrow (1), Gun (2), Knife (3), Lathi (4), Pistol (5), and Sword (6) are collected from differently publicly available object detection datasets such as ImageNet object detection dataset [2], ImageNet Large Scale Visual Recognition Challenge (ILSVRC) dataset [19] and Open Image Dataset V4 [14], from Internet using web scrapping, and some own captured images. Subsequently, the images are labeled using Image Labeling software. This "Weapon7" dataset comprises a total 17,935 number of images. These images are segregated as train set ($70\% \; of \; 17935 \approx 12551$), validation set ($20\% \; of \; 17935 \approx 3589$), and test set ($10\% \; of \; 17935 \approx 1795$). Subsequently, around 300 number of background images (having no trained objects) are added to the training set to reduce the false positives. The inherent data imbalance issue is addressed by giving more importance to the under-represented class and penalizing the over-represented class during training by adjusting the individual class weights in the configuration file according to the corresponding class distribution.

5.3 Performance evaluation metrics

In this weapon detection task, widely used performance metrics such as Precision (P), Recall (R), and mean Average Precision (mAP) are used to evaluate the performances of the proposed model. The Average Precision (AP) is calculated as the integral of the Precision-Recall (P-R) curve, with recall represented on the x-axis and precision represented on the y-axis as expressed in Eq. 3 [24].

$$AP = \int_{i=0}^{i=1} P_i R_i dR_i \tag{3}$$

Subsequently, mean Average Precision (mAP) for multiple classes is calculated by averaging the AP values of the classes as expresses in Eq. 4 [24].

$$mAP = \frac{1}{N} \sum_{c=0}^{N-1} AP_c \times 100\% \tag{4}$$

Here, $N = 7$ as there are 7 classes (#Cls) of the weapons such as Axe (0), Bow and arrow (1), Gun (2), Knife (3), Lathi (4), Pistol (5), and Sword (6). In case of object detection, mAP@0.5 is considered as the primary evaluation metric.

5.4 Implementation details and hyperparameters

The implementation details and hyper-parameters of the proposed YOLO-GTWDNet model are presented in Table 2. These values are selected based on comprehensive experimental analysis and hyperparameter tuning.

Table 1 Experimental setup

High-end system	Graphical computation system
CPU	Intel (R) Xeon (R) Silver 4214R CPU @ 2.40 GHz, 12 Core
GPU	NVIDIA RTX A4000, 16 GB
RAM	DDR4 (64 GB)
OS	UBUNTU 20.04 LTS (64 bits)
DL framework	Pytorch 2.0.1
Programming language	Python 3.10.11
CUDA compatibility	CUDA 11.7

Table 2 Hyperparameters of the proposed YOLO-GTWDNet model

Hyperparameter	Values
Optimizer	SGD
Learning rate	0.01
Momentum	0.0005
Warmup epoch	3
Warmup momentum	0.8
Warmup-bias learning rate	0.1
Patience for early stopping	50
Maximum epoch	500
Training completed at	144 epoch
Time taken for training completion	29.576 h

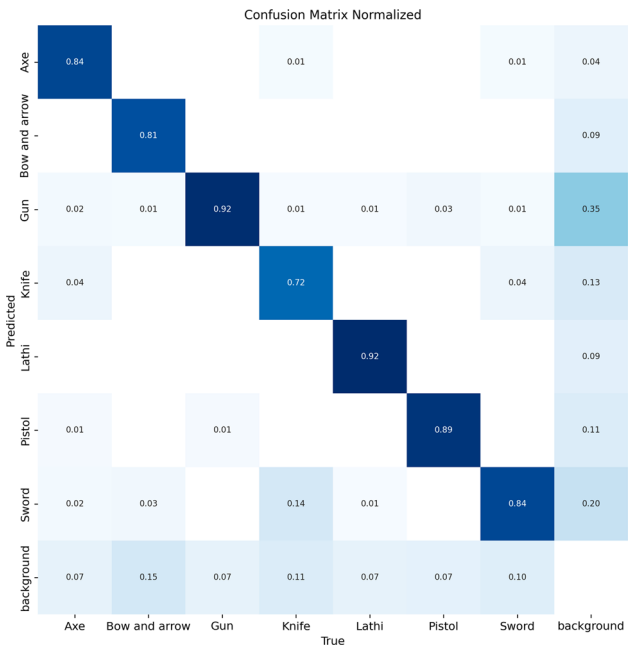

Fig. 3 Normalized confusion matrix of the proposed YOLO-GTWDNet model

5.5 Quantitative results analysis

Various quantitative performance metrics such as confusion matrix, F_1 score, and PR curve of the proposed YOLO-GTWDNet model for the test image set are presented in Fig. 3, 4 and 5, respectively. From these figures, it can be observed that the performance of the proposed model is quite promising. The performance of the proposed model is best for the gun class and worst for the knife class. This is primarily due to the very small size of the knife objects and the high structural similarity among knives, lathi, and swords.

An ablation study of the proposed YOLO-GTWDNet model is presented in Table 3 to systematically investigate the quantitative results for validation and test data splits. The model performances are found to be better as compared to

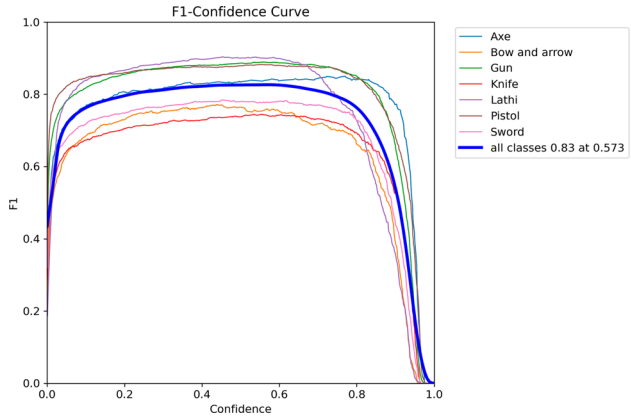

Fig. 4 F_1 curve

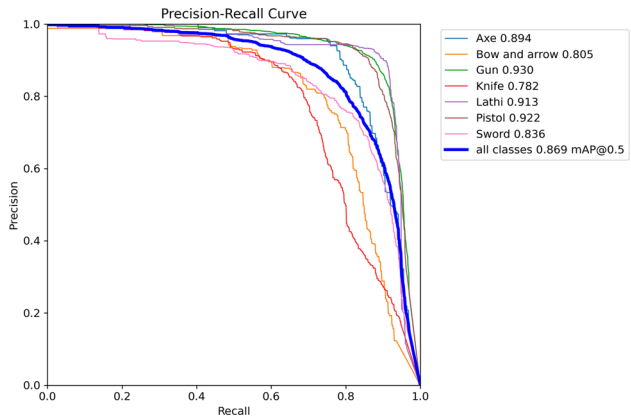

Fig. 5 PR curve

that of YOLOv8x, YOLOv8x-C3TR, YOLOv8x-GhostNet-SPPF-Bottlenck (YOLOv8x model with GhostNet backbone and SPPF bottleneck), and YOLOV8x-GhostNet-Bottleneck (YOLOv8x model with GhostNet backbone and Ghost bottleneck). Further, the proposed YOLO-GTWDNet model is compared with the base models such as YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x as shown in Table 3. Finally, this Table shows that the proposed model outperforms the base models and the other three hybrid models.

5.6 Runtime and computational complexity analysis

The runtime in terms of model speed and model complexities in terms of number of parameters, GFLOPS as well as FPS are presented in Table 4. The proposed YOLO-GTWDNet model provides better speed and FPS with fewer parameters and fewer GFLOPS. The proposed model is of more depth and lesser width to extract selectively better features.

Table 3 Ablation study of the proposed YOLO-GTWDNet model

Model	Test samples	Precision ↑	Recall ↑	mAP_{50} ↑	mAP_{50-95} ↑
YOLOv8n	Val	0.828	0.737	0.788	0.595
	Test	0.819	0.734	0.785	0.592
YOLOv8s	Val	0.835	0.761	0.836	0.627
	Test	0.832	0.758	0.824	0.625
YOLOv8m	Val	0.837	0.781	0.845	0.656
	Test	0.833	0.779	0.842	0.654
YOLOv8l	Val	0.859	0.795	0.865	0.687
	Test	0.857	0.792	0.856	0.685
YOLOv8x	Val	0.850	0.813	0.879	0.691
	Test	0.847	0.808	0.864	0.689
YOLOv8x-C3TR	Val	0.839	0.799	0.864	0.681
	Test	0.837	0.786	0.841	0.668
YOLOv8x-GhostNet-SPPF-Bottleneck	Val	0.849	0.761	0.829	0.639
	Test	0.847	0.758	0.822	0.636
YOLOv8x-GhostNet-GhostBottleneck	Val	0.836	0.765	0.839	0.657
	Test	0.834	0.764	0.837	0.654
YOLO-GTWDNet(ProposedModel)	Val	**0.863**	**0.811**	**0.879**	**0.695**
	Test	**0.860**	**0.797**	**0.869**	**0.693**

The bold value indicates the results of the proposed model and the best performances

Table 4 Computational complexity comparison for the proposed YOLO-GTWDNet model

Model	# Layers	#Parameters ↓	GFLOPs ↓	Model size (in MB) ↓	Model execution (in mS) ↓	FPS ↑
YOLOv8n	168	3012213	8.201	6.2	0.975	1025
YOLOv8s	168	11138309	28.66	22.5	1.917	521
YOLOv8m	218	25860373	79.085	52.0	4.108	243
YOLOv8l	268	43635237	165.419	87.6	6.854	145
YOLOv8x	268	68159349	258.155	136.7	12.227	81
YOLOv8x-C3TR	373	64882869	254.544	130.1	17.105	58
YOLOv8x-GhostNet-SPPF-Bottleneck	635	40583689	144.297	81.6	9.741	102
YOLOv8x-GhostNet-GhostBottleneck	639	43052969	147.261	86.6	9.598	104
YOLO-GTWDNet (Proposed Model)	**647**	**39776489**	**143.651**	**80.0**	**8.588**	**116**

The bold value indicates the results of the proposed model and the best performances

5.7 Qualitative result analysis

The detection results of the proposed YOLO-GTWDNet model in terms of class names with associated confidence scores for given test images are presented in Fig. 6. It is evident from this qualitative result that lathi can be detected in the crowded and partially occluded scenario.

5.8 Comparative analysis

The proposed YOLO-GTWDNet outperforms the SOTA methods as mentioned in the Table 5. The proposed model provides competitive detection accuracy with higher FPS.

(a) Street scenario (b) Urban scenario

Fig. 6 Detection results for typical test image

Table 5 Comparative analysis

Method	#Cls	P	R	mAP	FPS
[9]	5	N/A	N/A	0.740	N/A
[7]	2	0.840	0.710	0.773	N/A
[8]	1	0.793	0.686	0.540	N/A
Proposed YOLO-GTWDNet	**7**	**0.860**	**0.797**	**0.869**	**116**

The bold value indicates the results of the proposed model and the best performances

6 Conclusion

In the rapidly evolving landscape of smart cities, ensuring public safety remains a paramount concern. Hence, an efficient, novel, robust, and lightweight YOLO-GTWDNet model is proposed for detecting the weapons from stored images or live video streams. The proposed model achieves lightweight by introducing GhostNet in the backbone of the YOLOv8x model. The accuracy is further enhanced by introducing GhostNet bottleneck and C3TR in the neck section of the YOLOv8x model. Comprehensive experimental analysis validated the effectiveness of the proposed YOLO-GTWDNet model. The proposed model can be deployed in IVSS for smart city applications. Future work may be based on the extension of this work with more novel variants of the model architecture to improve accuracy further without compromising on the inference speed.

Acknowledgements This work was supported by the IMPRINT (Grant No.-7794/2016), an initiative of Ministry of Education, Government of India and Ministry of Housing and Urban Affairs, Government of India.

Author Contributions R. Nayak did the work and wrote the manuscript. U. C. Pati and S. K. Das supervised the work and provided the resources. G. K. Sahoo prepared the data set along with R. Nayak and executed the preprocessing of the dataset. All the authors reviewed the manuscript.

Data availability The authors confirm that the data supporting the findings are collected from ImageNet object detection dataset [2], ImageNet Large Scale Visual Recognition Challenge (ILSVRC) dataset [19] and Open Image Dataset V4 [14] and some own captured images. The publicly available repositories can be accessed. The said databases can be accessed on web page https://www.image-net.org/challenges/LSVRC/index.php and https://storage.googleapis.com/openimages/web/index.html. These databases were last accessed by the authors for the implementation used on 5th Jun 2023. The complete Weapon7 dataset can be shared upon individual requests.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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